Pre-Processing of Remotely Sensed Data

Pre-Processing

- Raw RS data may contain flaws and deficiencies, the corrections and removal of flaws present in the data is termed as Pre-processing
- Pre-processing consists of those operations that prepare data for subsequent analysis
- It is a preparatory phase that in principle improves image quality as the basis for latter analysis

Pre-Processing

- No definitive list of standard preprocessing steps for each application/data
- Quality of image data vary greatly; some data may not require any pre-processing that would be necessary in other instances
- Pre-Processing may change data, add unwanted effects upon or alter digital values

Pre-Processing

- **Must consider that these changes will be beneficial for later analysis, some change may not immediately obvious, so analyst should decide what pre-processing is essential for the application data would be used for , or to obtain specific results**

IMAGE ENHANCEMENT

Image Enhancement

- What is image enhancement?
- Different kinds of image enhancement
- Histogram processing
- Point processing
- Neighbourhood operations

IMAGE ENHANCEMENT

- The alteration of the appearance of an image in such a way that the info contained in that image is more readily interpreted visually in terms of a particular need
- It alters the visual impact of the image to improves the info contents for the interpreter

What Is Image Enhancement?

- **Image enhancement is the process of making images more useful thereby**
  - **Highlighting interesting detail in images**
  - **Removing noise from images**
  - **Making images more visually appealing**

IMAGE ENHANCEMENT

- **No single standard method can be said to be the best, it depends upon the need of the user**
- **The characteristic of each image in terms of distribution of pixel values over (0-255 range) will change from one area to another , thus enhancement tech suited for one image may not be good for other image covering different type of area**

Image Histogram

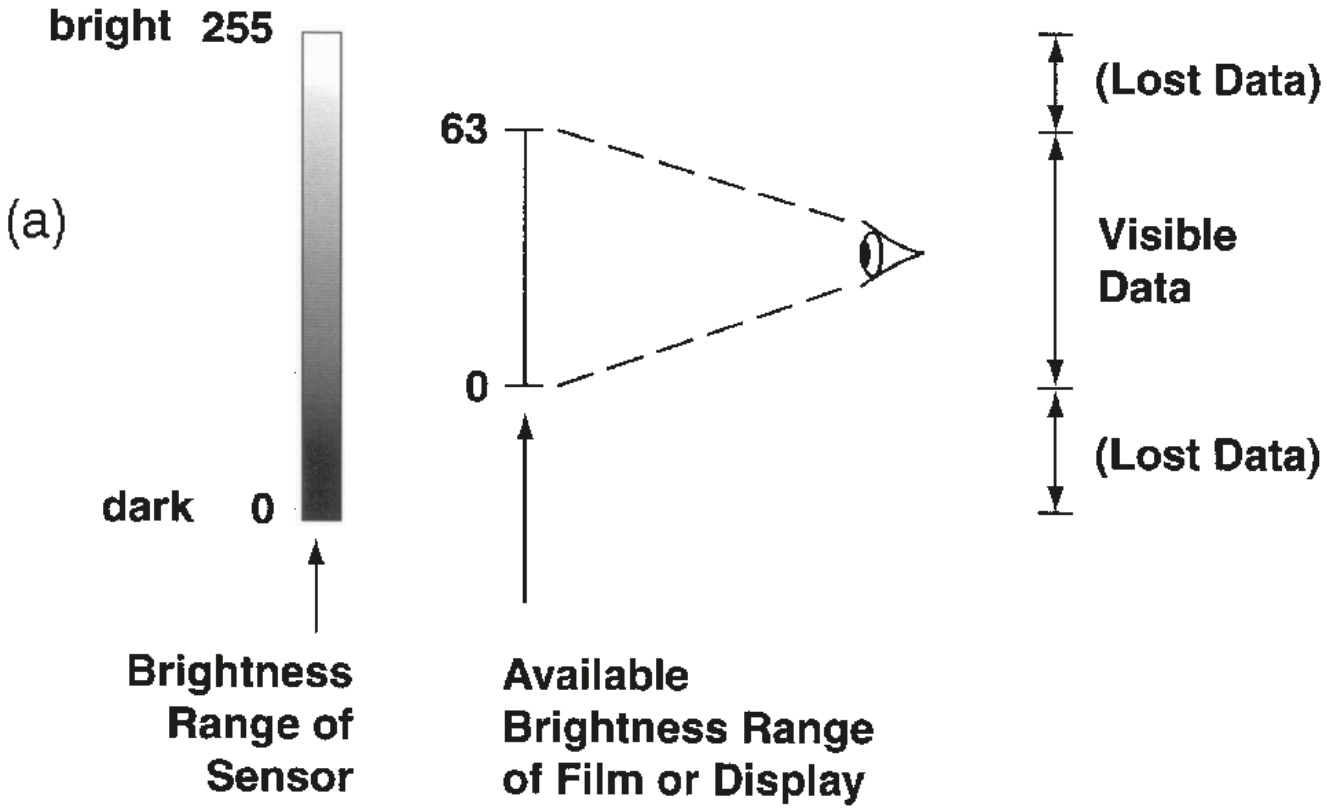
- Histogram greatly helps to assess the appearance of an image
  - In a dark image, the gray levels would be clustered towards the lower end
  - In a bright image, the gray levels would be clustered towards the upper end
  - In a well contrasted image, the gray levels would be well spread out over much of the range

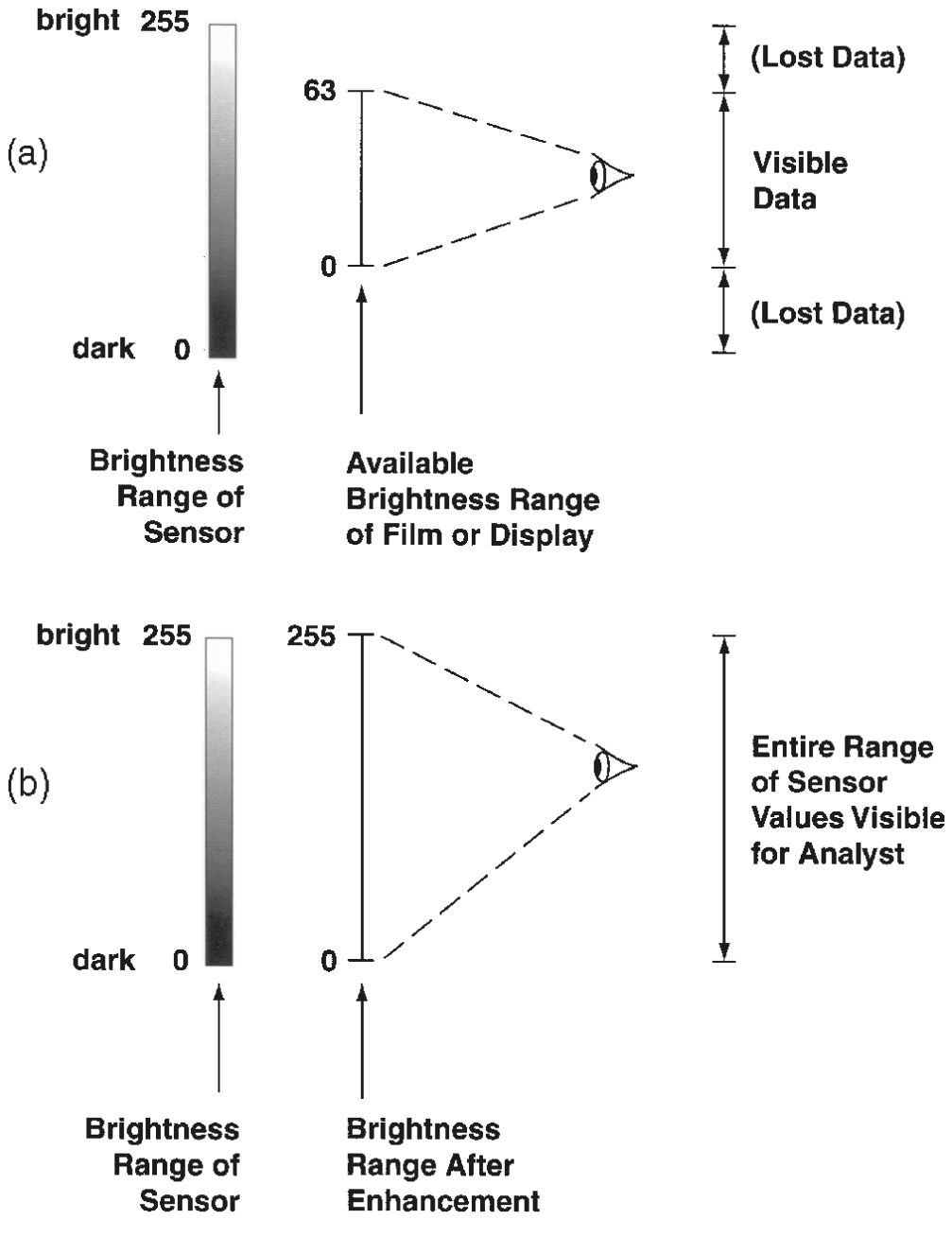
CONTRAST ENHANCEMENT/ STRETCHING

- Ideally one material would reflect tremendous amount of energy in a certain wavelength while another much less in the same wavelength
- This would result in contrast between the two types of materials
- In some cases, different materials would often reflect similar amount of radiant flux throughout the visible and IR portion of EM spectrum resulting in a relatively low contrast image

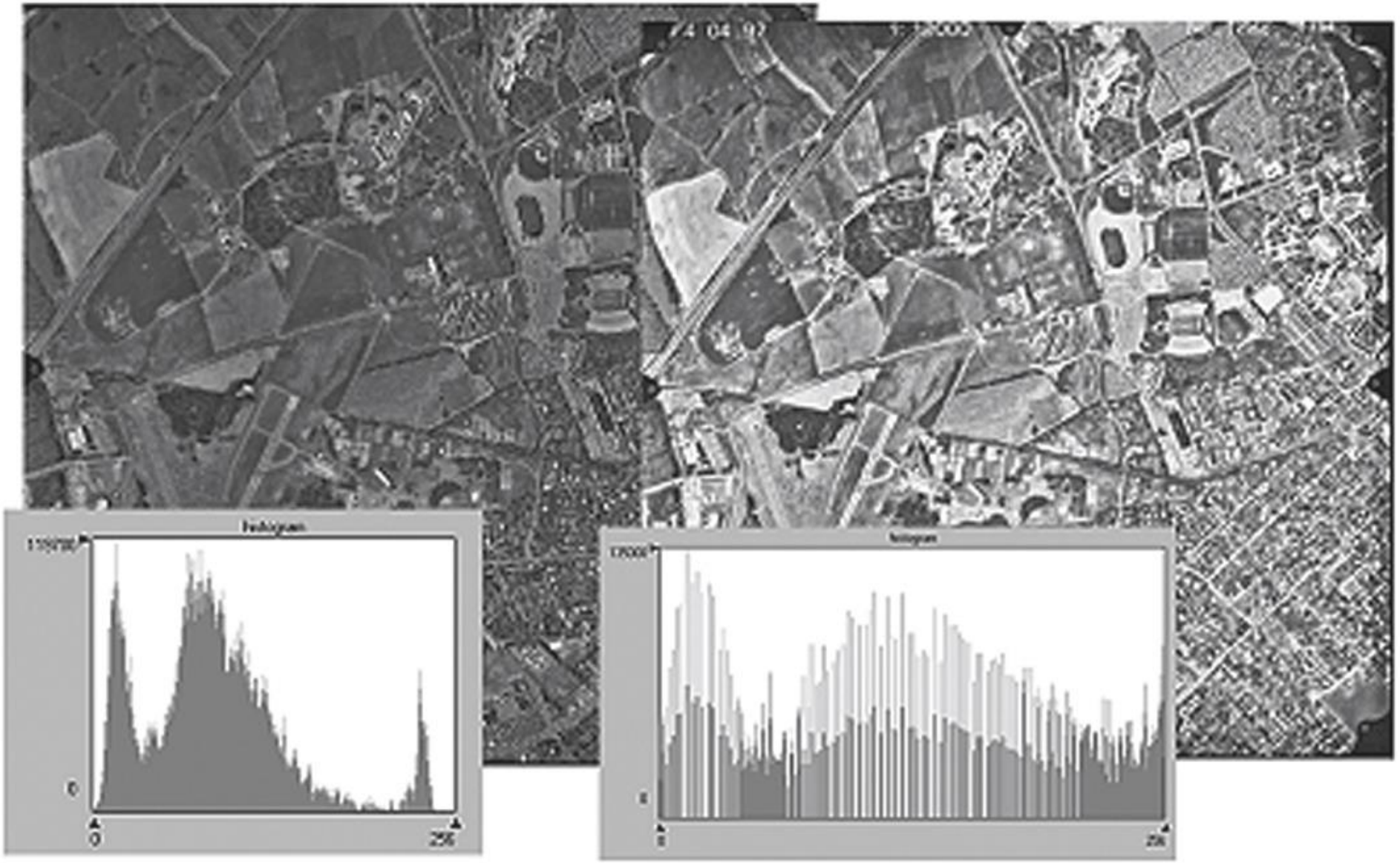
CONTRAST ENHANCEMENT

- **For area to be imaged, it is unlikely that full dynamic range (radiometric resolution) of the sensor will be used, thus the corresponding image is dull or over bright-over or under exposed**


bright 255
(a)
dark 0
Brightness Range of Sensor
63
0
Available Brightness Range of Film or Display
(Lost Data)
Visible Data
(Lost Data)


bright 255
(a)
dark 0
Brightness Range of Sensor
63
0
Available Brightness Range of Film or Display
(Lost Data)
Visible Data
(Lost Data)
bright 255
(b)
dark 0
Brightness Range of Sensor
255
0
Brightness Range After Enhancement
Entire Range of Sensor Values Visible for Analyst

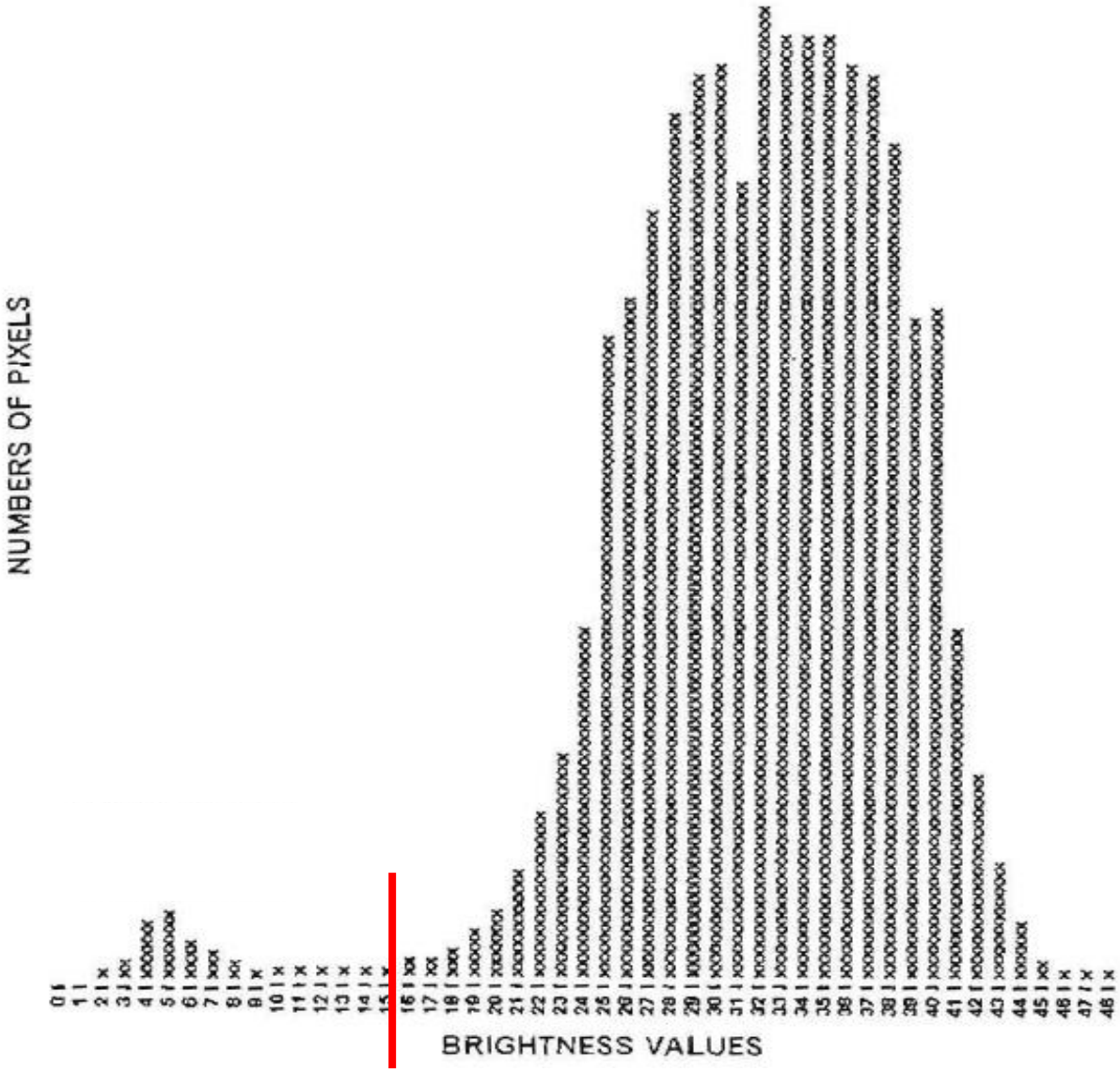
CONTRAST ENHANCEMENT



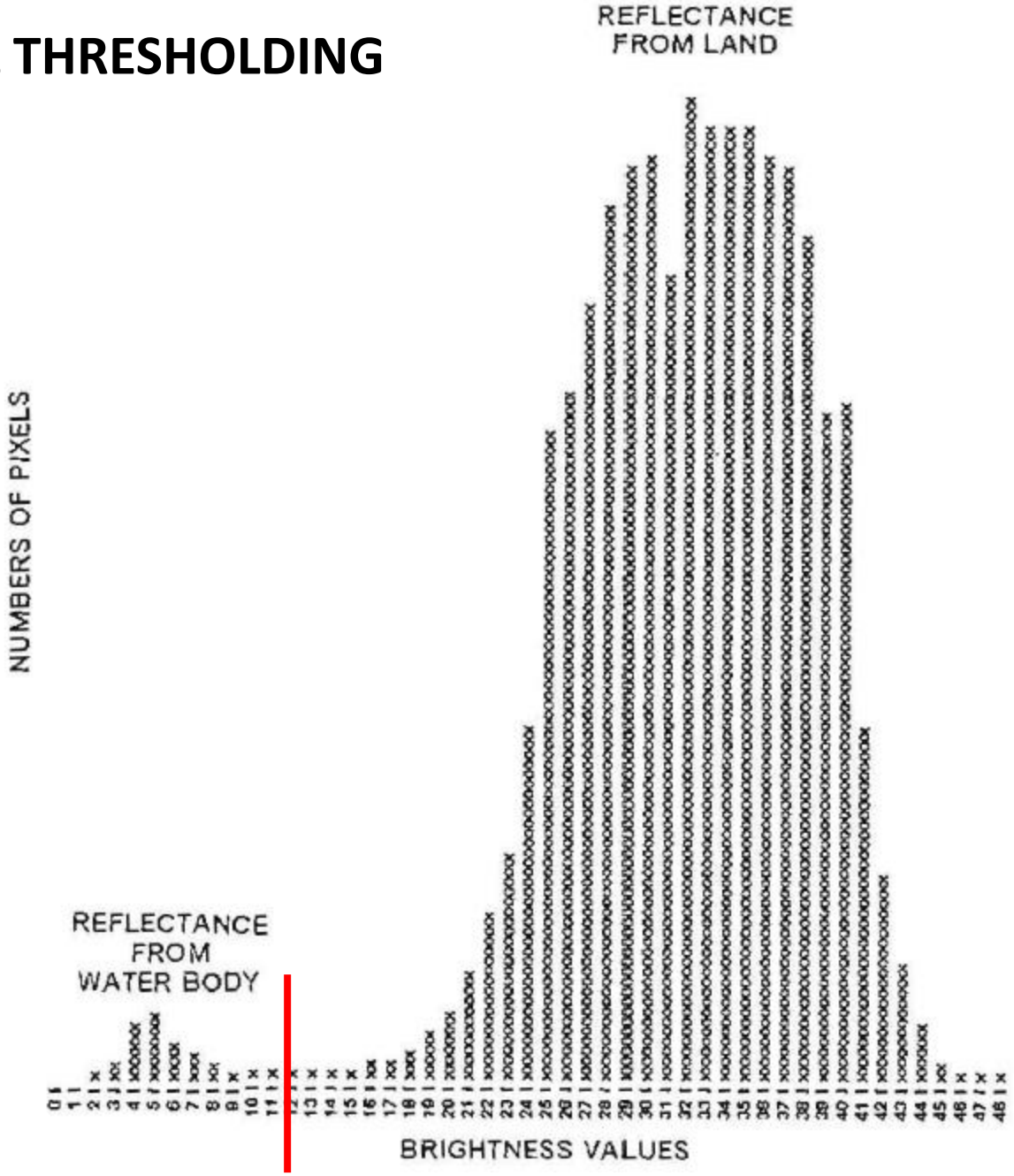
TYPES

- **CONTRAST ENHANCEMENT**
  - **GRAY LEVEL THRESHOLDING – Two classes (or more) classes**
  - **CONTRAST STRETCHING**

GRAY LEVEL THRESHOLDING



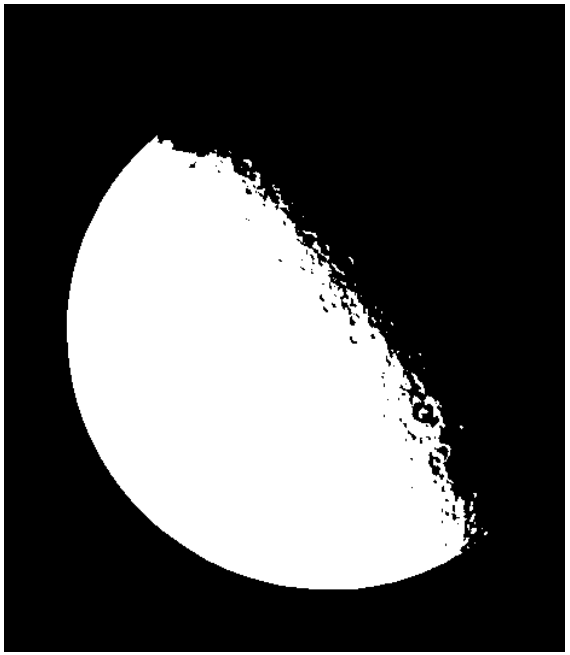
GRAY LEVEL THRESHOLDING



Thresholding

- **Thresholding transformations are particularly useful to isolate an object of interest from a background**





Grey Level Slicing

- **Highlights a specific range of grey levels**
  - **Like thresholding**
  - **Other levels can be suppressed or maintained**
  - **Useful for highlighting features in an image**

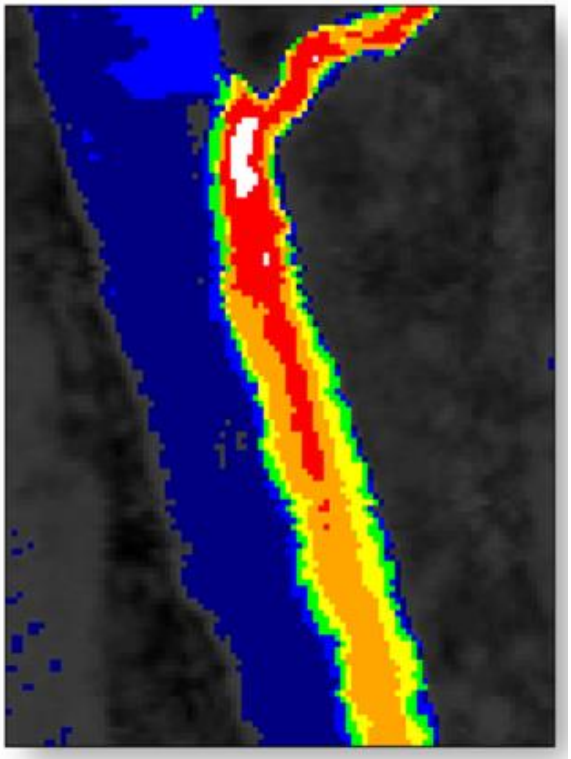
Density/ Level Slicing

- **Representation of a range of contiguous gray levels of gray scale image by a single color (Color ramp)**
- **Used to separate the data into “n” intervals or “slices” based on the histogram from one wavelength band**
- **All data within a slice are displayed as one digital number or color in the output image**
- **The gray levels in the output image corresponds to the number of slices**

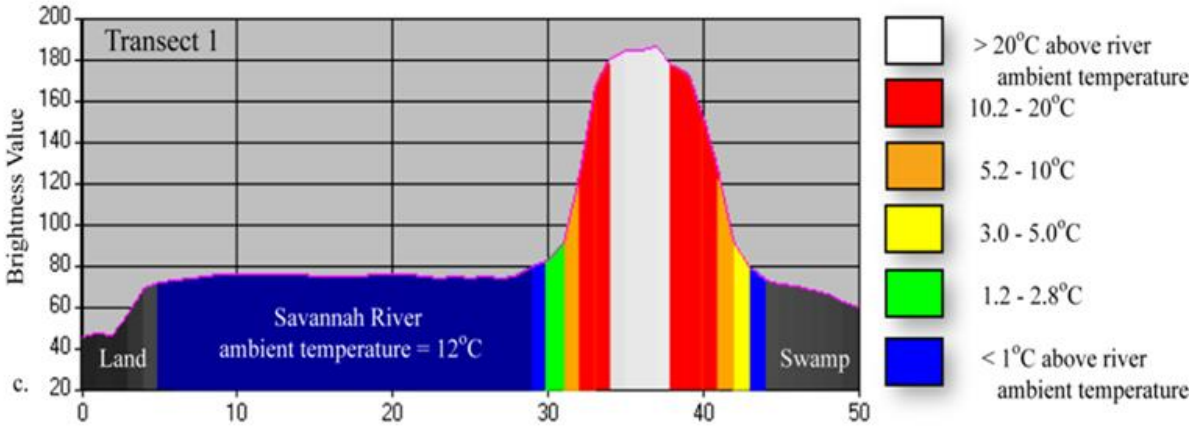
Density/ Level Slicing

- **Used frequently with thermal images, i.e., different temperature ranges can be shown with different slices, for Example**
  - **Split temp range to some xx°C intervals**
  - **Each interval represented by a color, such that temperature information in the image can be analyzed more easily because the differences between the colors are greater than the black and white**

Density Slicing

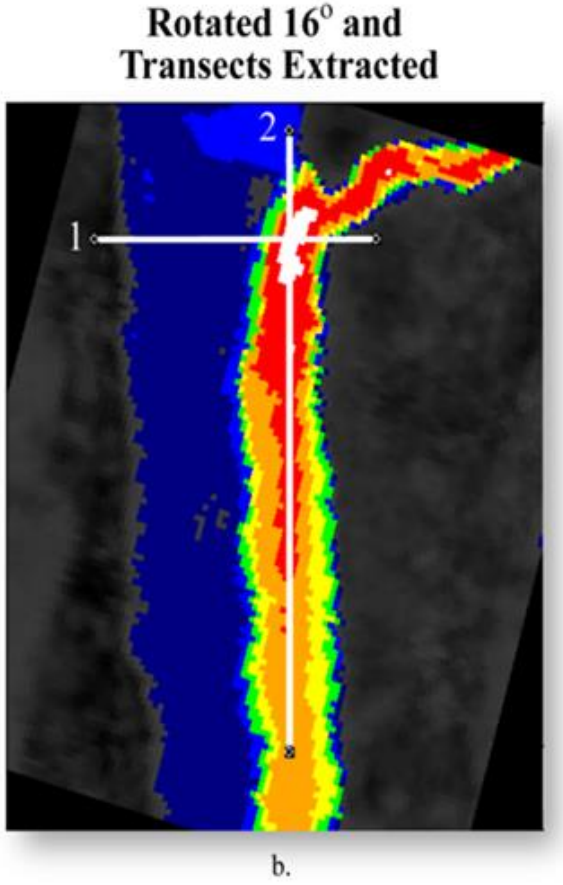


a.

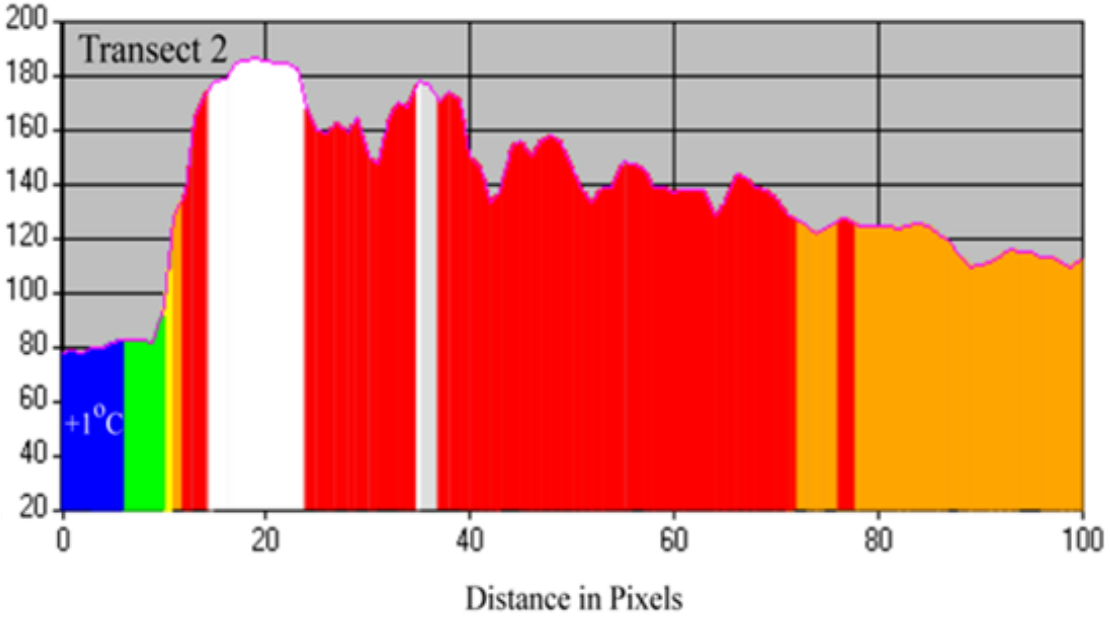


c.

Density Slicing



b.



LINEAR STRETCH-MIN MAX

The uniform expansion of the of input digital numbers to full range)-255) is called linear stretch

$$BV_{OUT} = 255(BV_{IN} - MIN_k) / (MAX_k - MIN_k)$$

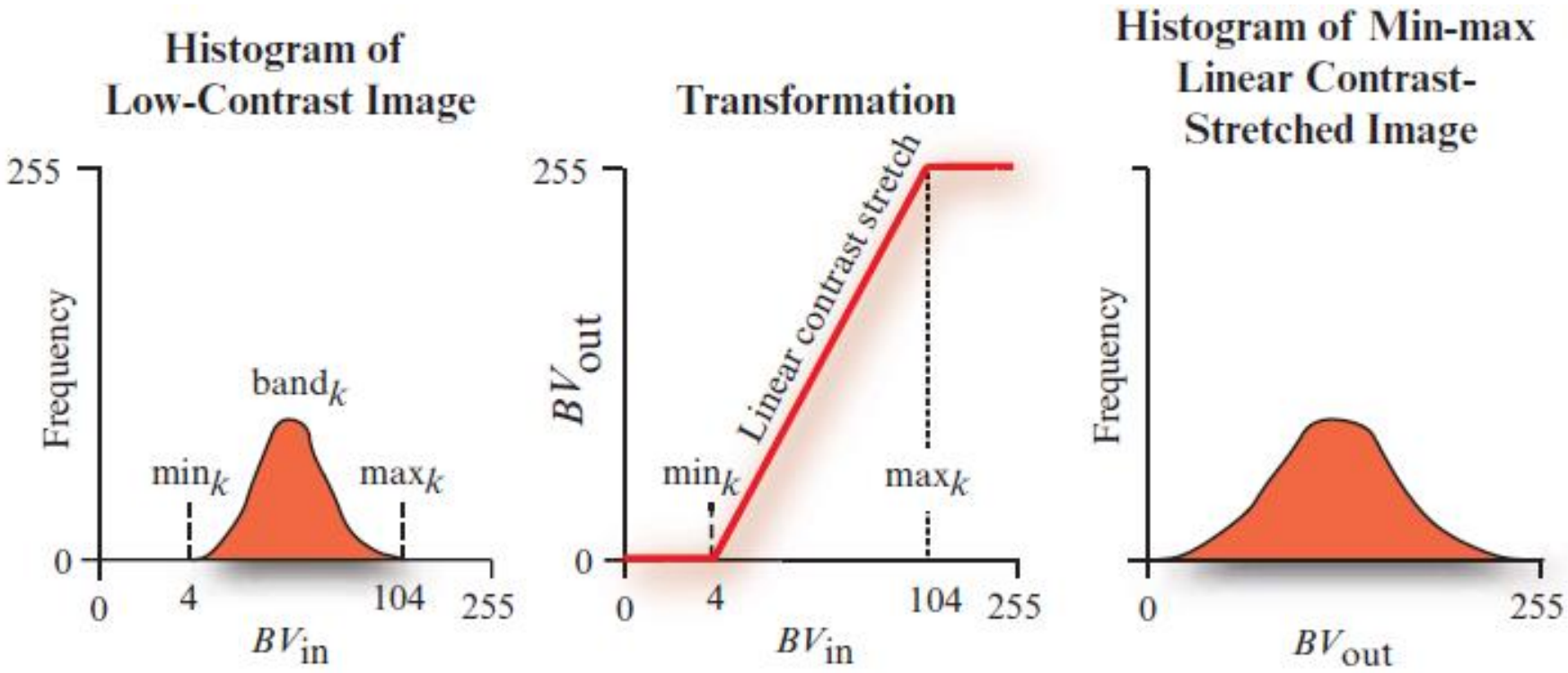
MIN=25, MAX=225

130	165	100	135	145
57	35	25	50	65
180	215	135	200	205
205	205	225	220	225
30	25	25	120	205



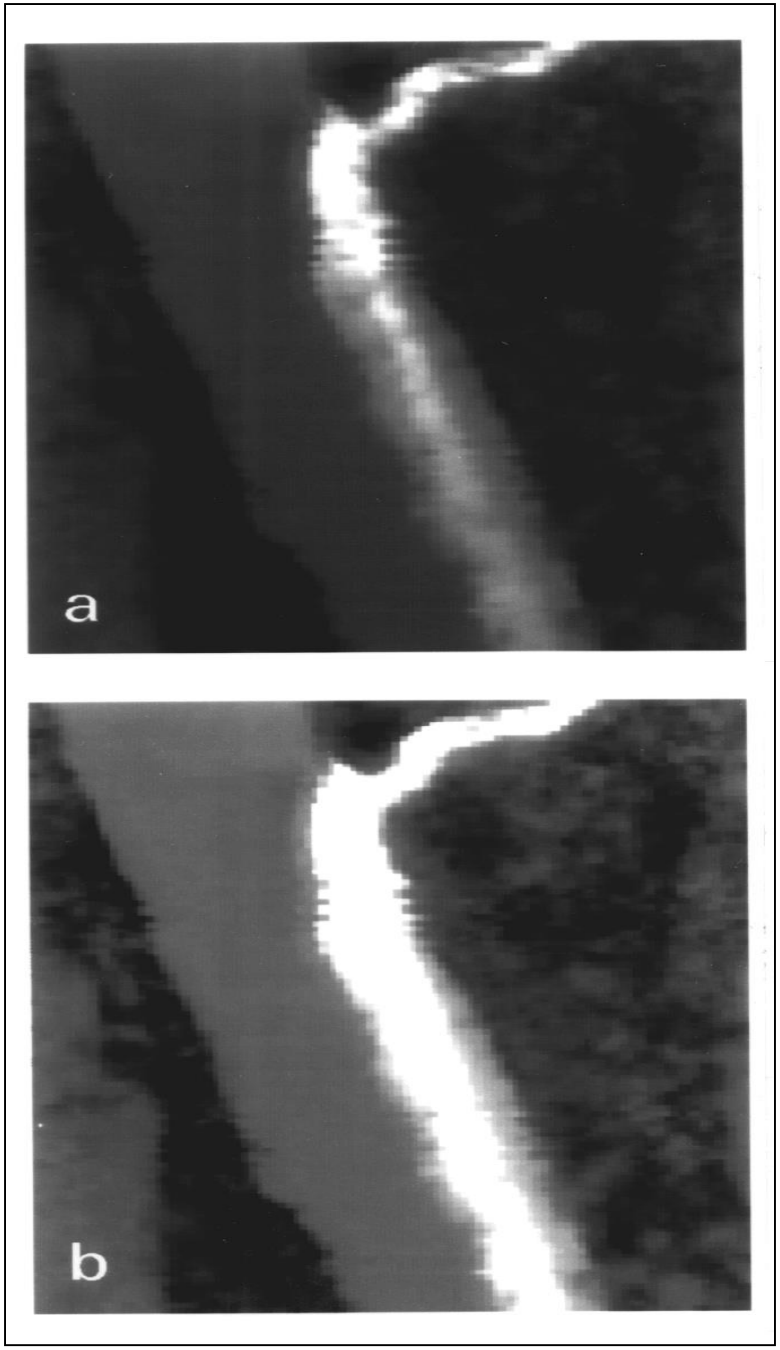
134	179	96	140	153
41	13	0	32	51

Min-Max Contrast Stretch

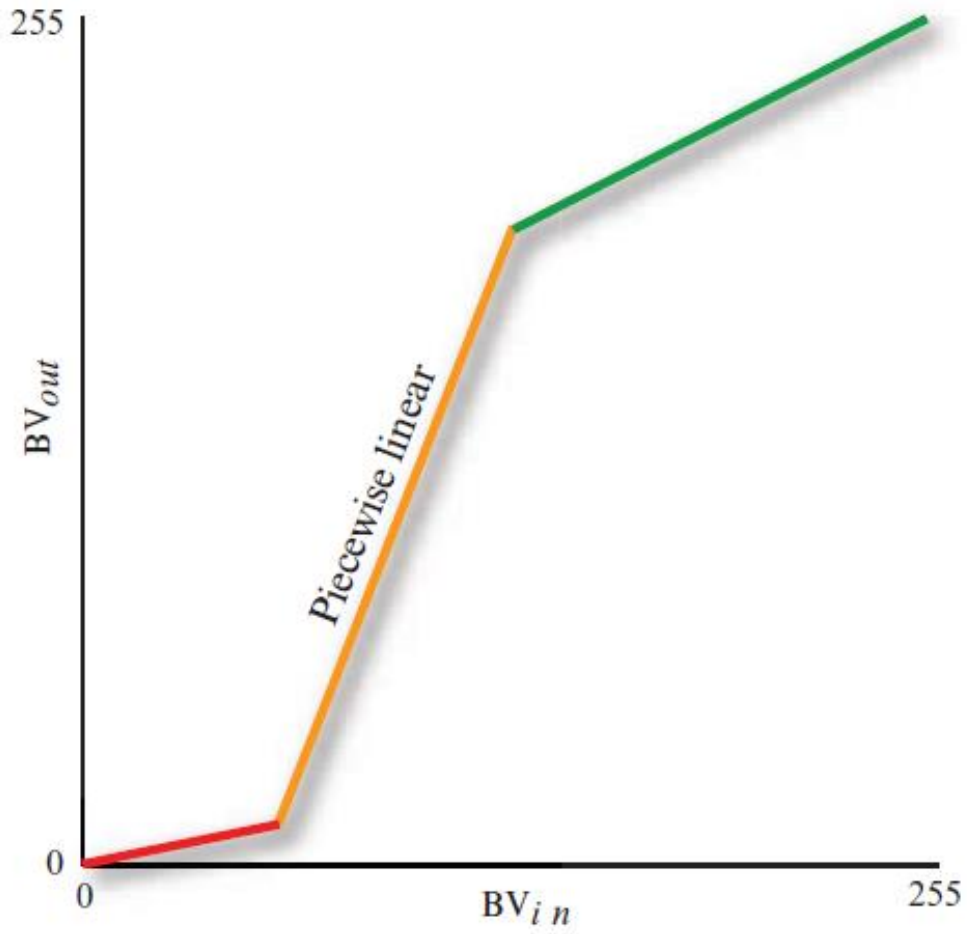


Original

Minimum-maximum Stretch



Piecewise linear Contrast Stretching

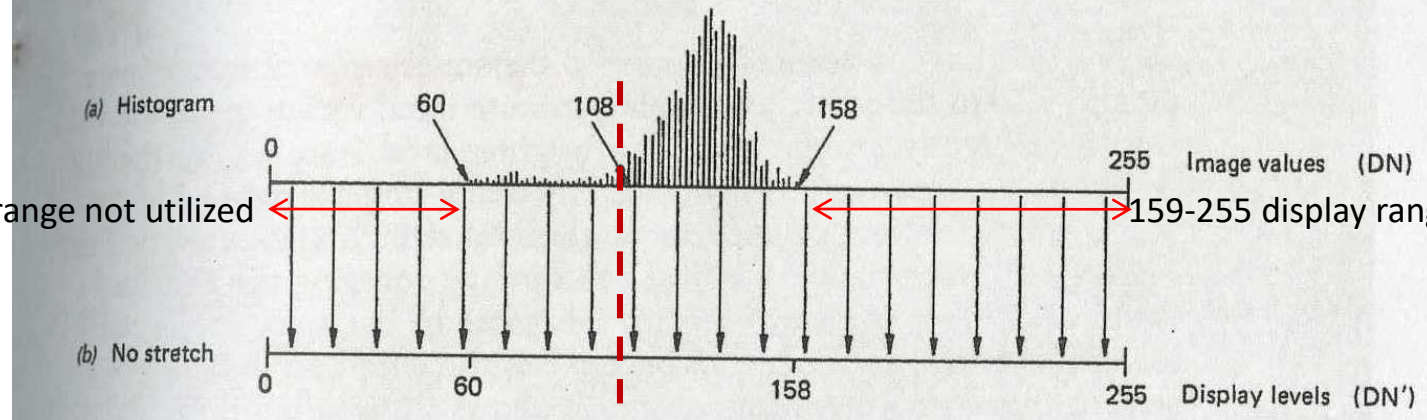


- *The original brightness range is divided into segments before* each segment is stretched individually.
- This variation permits the analyst to emphasize certain segments of the brightness range (setting up a series of min_k and max_k*)* that might have more significance for a specific application
- When histogram of the image is bimodal, tri-modal, It is possible to perform piecewise linear Contrast stretching

HISTOGRAM CONVERSION

- The histogram of the original image is converted to other types of histograms as specified by user
  - **Histogram Stretch-** Image values are assigned to the display levels based on their frequency of occurrence
  - More display values (more radiometric details) are assigned to the frequently occurring portion of histogram
  - **Special Stretch-** To analyze specific features in greater radiometric details by assigning the display range exclusively to a particular range of image values

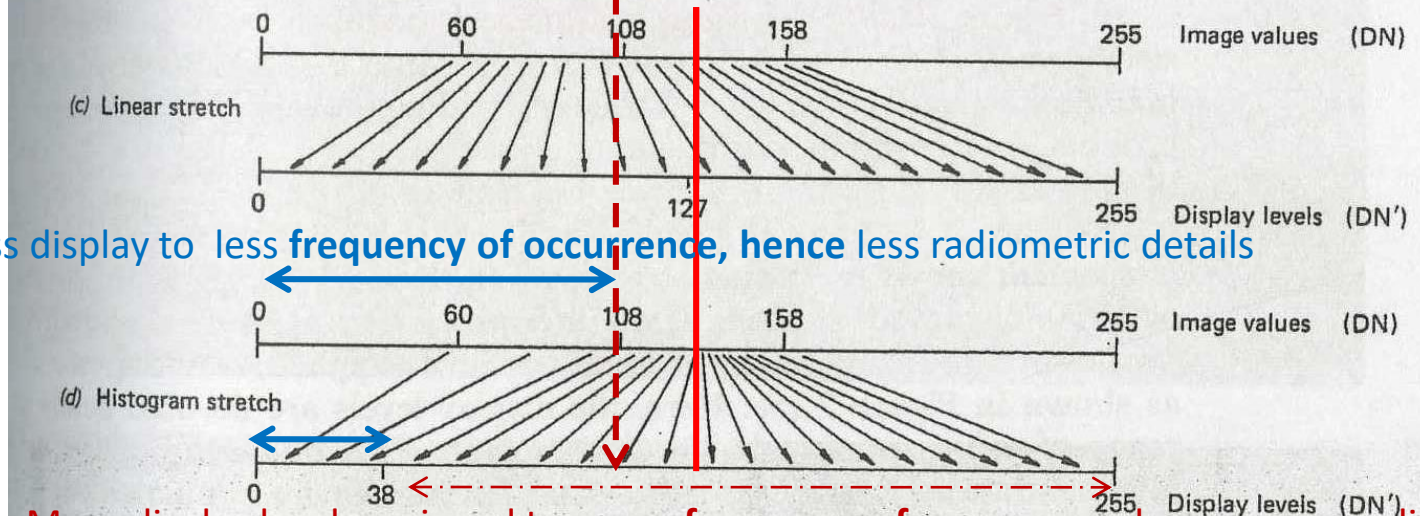
Tonal info is compressed to small range , reducing the interpretation of radiometric details



0-59 display range not utilized

159-255 display range not utilized

Expanding the tonal levels to fill the full range of display, will not provide most expressive display



Less display to less **frequency of occurrence, hence** less radiometric details

More display levels assigned to more **frequency of occurrence**, hence more radiometric details

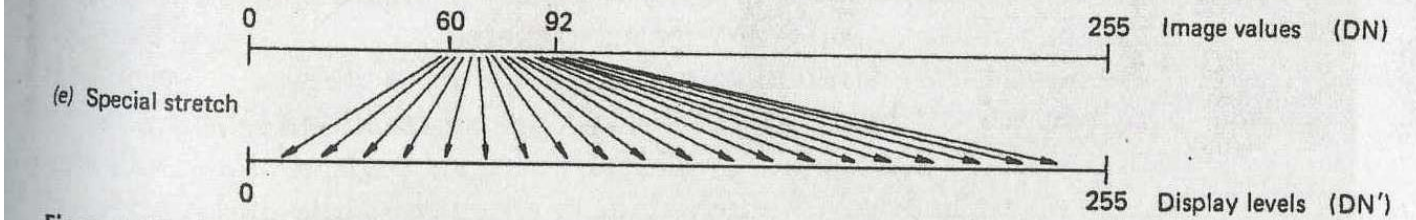
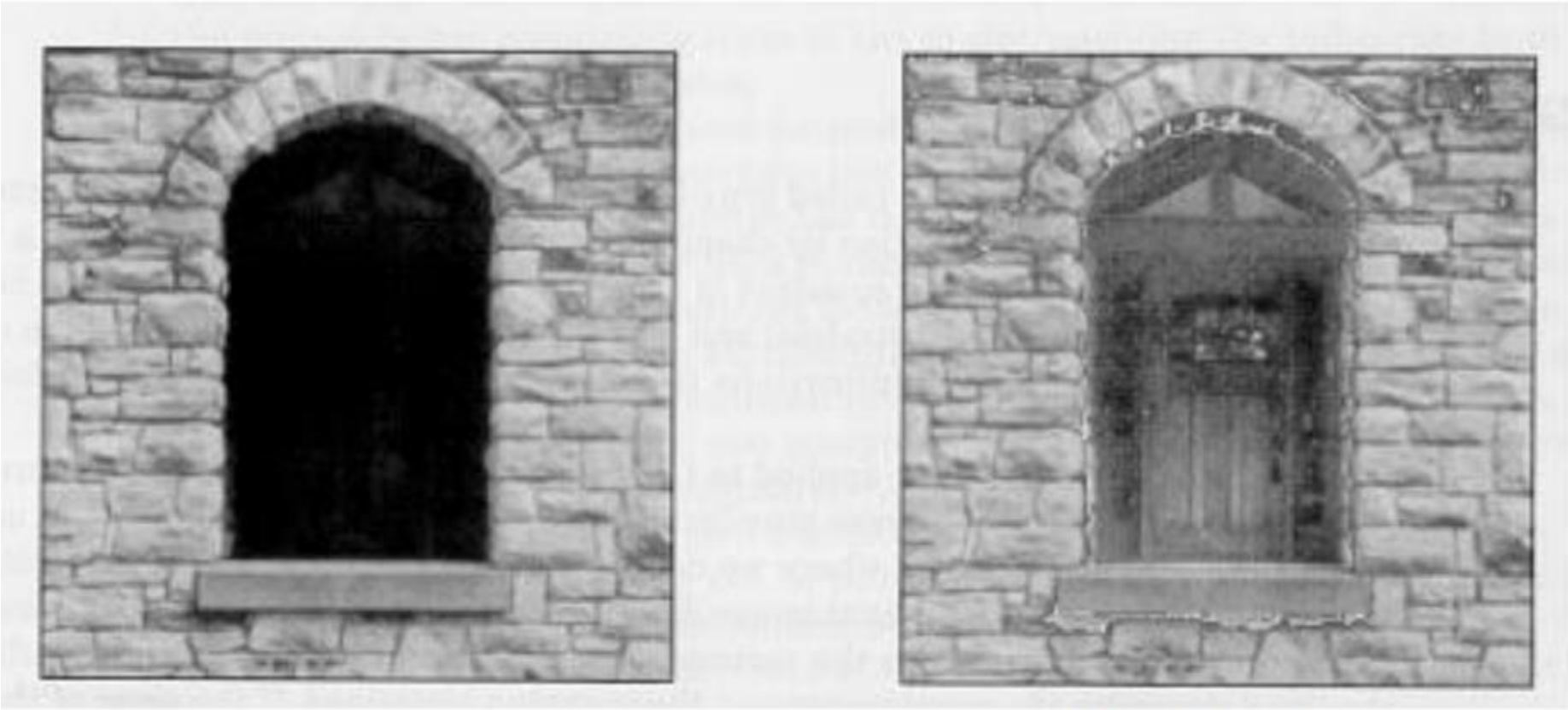


Figure 7.13 Principle of contrast stretch enhancement.

Contrast Enhancement



Special Stretch



Before Stretch

After Stretch

CONTRAST ENHANCEMENT

- If the range of gray levels could be altered so as to fit the full range of the black and white axis, then the contrast between the dark and bright areas of the image would be improved
- Does not modify the original data unless new file is saved

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